Lesson 4: Integration by Trigonometric Substitution

Criteria	Exceptional (4)	Proficient (3)	Basic (2)	Needs Improvement (1)
Identification & Setup	Correctly identifies form ($\sqrt{a^2 - x^2}$, $\sqrt{a^2 + x^2}$, $\sqrt{x^2 - a^2}$) and selects the correct substitution ($a\sin\theta$, $a\tan\theta$, $a\sec\theta$).	Correct form identified, but minor error in substitution constants (a and b).	Incorrect substitution selected, making the integral harder to solve.	Unable to determine that trig substitution is needed.
Substitution & Differential	Correctly substitutes x and dx (including dθ) and simplifies the radical using Pythagorean identities ($\sin^2\theta + \cos^2\theta = 1$)	Substitutes x and dx correctly, but has minor errors in algebraic simplification of the radical.	Correct trig substitution but fails to substitute dx correctly.	Incorrect substitution and differential leading to complex, unsolvable expressions.

Evaluation of Integral	Evaluates the trigonometric integral accurately, using necessary identities (e.g., half-angle) and techniques.	Integrates correctly, but makes a minor error in trigonometric simplification or integration rules.	Demonstrates a valid method, but errors in integration result in a wrong answer.	Unable to evaluate the simplified trigonometric integral.
Back-Substitution	Correctly uses a right triangle to express the result in terms of x and includes the constant of integration (+C).	Uses a triangle to convert back, but makes a minor error in substituting x expressions.	Attempts to convert back to x but uses wrong trigonometric ratios.	Final answer left in terms of θ. .
Definite Integrals	Correctly transforms limits of integration and evaluates the definitive value.	Transforms limits but makes errors in final evaluation.	Fails to change limits to θ and fails to convert back to x for evaluation.	No attempt to change limits or convert back.